Memory

A memory is just like a human brain. It is used to store data and instructions. Computer memory is the storage space in the computer, where data is to be processed and instructions required for processing are stored. The memory is divided into large number of small parts called cells. Each location or cell has a unique address, which varies from zero to memory size minus one. For example, if the computer has 64k words, then this memory unit has 64 * 1024 = 65536 memory locations. The address of these locations varies from 0 to 65535.

Memory is primarily of three types −

- Cache Memory
- Primary Memory/Main Memory
- Secondary Memory

a) Cache Memory

Cache memory is a very high speed semiconductor memory which can speed up the CPU. It acts as a buffer between the CPU and the main memory. It is used to hold those parts of data and program which are most frequently used by the CPU. The parts of data and programs are transferred from the disk to cache memory by the operating system, from where the CPU can access them.

The advantages of cache memory are as follows −

- Cache memory is faster than main memory.
- It consumes less access time as compared to main memory.
- It stores the program that can be executed within a short period of time.
- It stores data for temporary use.

The disadvantages of cache memory are as follows −

- Cache memory has limited capacity.
- It is very expensive.

b) Primary Memory / Main Memory

Primary memory holds only those data and instructions on which the computer is currently working. It has a limited capacity and data is lost when power is switched off. It is generally made up of semiconductor device. These memories are not as fast as registers. The data and instruction required to be processed resides in the main memory. It is divided into two subcategories RAM and ROM.

Characteristics of Main Memory

- These are semiconductor memories.
- It is known as the main memory.
- Usually volatile memory.
- Data is lost in case power is switched off.
- It is the working memory of the computer.
- Faster than secondary memories.
- A computer cannot run without the primary memory.

c) Secondary Memory

This type of memory is also known as external memory or non-volatile. It is slower than the main memory. These are used for storing data/information permanently. CPU directly does not access these memories, instead they are accessed via input-output routines. The contents of secondary memories are first transferred to the main memory, and then the CPU can access it. For example, disk, CD-ROM, DVD, etc.

Characteristics of Secondary Memory

- These are magnetic and optical memories.
- It is known as the backup memory.
- It is a non-volatile memory.
- Data is permanently stored even if power is switched off.
- It is used for storage of data in a computer.
- Computer may run without the secondary memory.
- Slower than primary memories.

Memory Devices

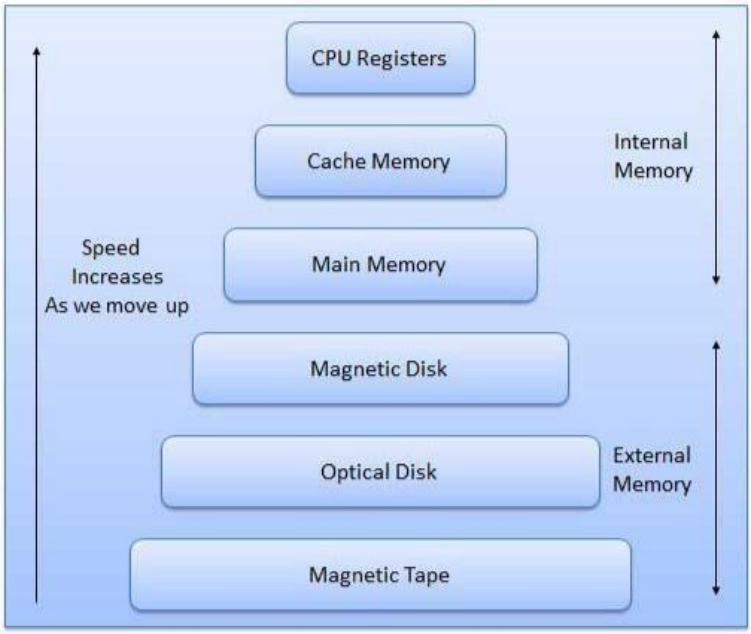
A memory is just like a human brain. It is used to store data and instruction. Computer memory is the storage space in computer where data is to be processed and instructions required for processing are stored.

The memory is divided into large number of small parts. Each part is called a cell. Each location or cell has a unique address which varies from zero to memory size minus one.

For example if computer has 64k words, then this memory unit has 64 * 1024 = 65536 memory location. The address of these locations varies from 0 to 65535.

Memory is primarily of two types

- Internal Memory − cache memory and primary/main memory
- External Memory − magnetic disk / optical disk etc.



Characteristics of Memory Hierarchy are following when we go from top to bottom.

- Capacity in terms of storage increases.
- Cost per bit of storage decreases.
- Frequency of access of the memory by the CPU decreases.
- Access time by the CPU increases.

RAM (Random Access Memory)

A RAM constitutes the internal memory of the CPU for storing data, program and program result. It is read/write memory. It is called random access memory(RAM).

Since access time in RAM is independent of the address to the word that is, each storage location inside the memory is as easy to reach as other location & takes the same amount of time. We can reach into the memory at random & extremely fast but can also be quite expensive.

RAM is volatile, i.e. data stored in it is lost when we switch off the computer or if there is a power failure. Hence, a backup uninterruptible power system(UPS) is often used with computers. RAM is small, both in terms of its physical size and in the amount of data it can hold.

RAM is of two types-

- Static RAM(SRAM)
- Dynamic RAM(DRAM)

Static RAM(SRAM)

The word static indicates that the memory retains its contents as long as power remains applied. However, data is lost when the power gets down due to volatile nature. SRAM chips use a matrix of 6-transistors and no capacitors. Transistors do not require power to prevent leakage, so SRAM need not have to be refreshed on a regular basis.

Because of the extra space in the matrix, SRAM uses more chips than DRAM for the same amount of storage space, thus making the manufacturing costs higher.

Static RAM is used as cache memory needs to be very fast and small.

Dynamic RAM(DRAM)

DRAM, unlike SRAM, must be continually refreshed in order for it to maintain the data. This is done by placing the memory on a refresh circuit that rewrites the data several hundred times per second. DRAM is used for most system memory because it is cheap and small. All DRAMs are made up of memory cells.

These cells are composed of one capacitor and one transistor.

The capacitor can be either charged or discharged; these two states are taken to represent the two values of a bit, conventionally called 0 and 1.

Transistors always leak a small amount, the capacitors will slowly discharge, and the information eventually fades unless the capacitor charge is refreshed periodically.

SRAM	DRAM
Stores data till the power is supplied	Stores data only for few milliseconds even when power is supplied
Uses an arry of 6 transistors for each memory cell	Uses a single transistor and capaitor for each memory cell
Does not refreshes the memory Cell	Needs to refresh the memory cell after each reading of the capacitor
Data access is faster	Data acces is slower
Consume more power	Consume less power
Low density/less memory per chip	High density/more memory per chip
Cost per bit is high	Cost per bit is low

ROM (Read Only Memory)

ROM stands for Read Only Memory. The memory from which we can only read but cannot write on it. This type of memory is non-volatile. The information is stored permanently in such memories during manufacture.

A ROM, stores such instruction as are required to start computer when electricity is first turned on, this operation is referred to as bootstrap.

ROM chip are not only used in the computer but also in other electronic items like washing machine and microwave oven.

Following are the various types of ROM −

a) MROM(Masked ROM)

The very first ROMs were hard-wired devices that contained a pre-programmed set of data or instructions. These kind of ROMs are known as masked ROMs. It is inexpensive ROM.

b) PROM(Programmable Read Only Memory)

PROM is read-only memory that can be modified only once by a user. The user buys a blank PROM and enters the desired contents using a PROM programmer. Inside the PROM chip there are small fuses which are burnt open during programming. It can be programmed only once and is not erasable.

c) EPROM(Erasableand Programmable Read Only Memory)

The EPROM can be erased by exposing it to ultra-violet light for a duration of upto 40 minutes. Usually, an EPROM eraser achieves this function. During programming an electrical charge is trapped in an insulated gate region. The charge is retained for more than ten years because the charge has no leakage path. For erasing this charge, ultra-violet light is passed through a quartz crystal window lid. This exposure to ultra-violet light dissipates the charge. During normal use the quartz lid is sealed with a sticker.

d) EEPROM(Electrically Erasable and Programmable Read Only Memory)

The EEPROM is programmed and erased electrically. It can be erased and reprogrammed about ten thousand times. Both erasing and programming take about 4 to 10 ms millisecond. In EEPROM, any location can be selectively erased and programmed. EEPROMs can be erased one byte at a time, rather than erasing the entire chip. Hence, the process of re-programming is flexible but slow.

Cache Memory

Cache memory is a very high speed semiconductor memory which can speed up CPU. It acts as a buffer between the CPU and main memory. It is used to hold those parts of data and program which are most frequently used by CPU. The parts of data and programs, are transferred from disk to cache memory by operating system, from where CPU can access them.

Advantages

- Cache memory is faster than main memory.
- It consumes less access time as compared to main memory.
- It stores the program that can be executed within a short period of time.
- It stores data for temporary use.

Disadvantages

- Cache memory has limited capacity.
- It is very expensive.

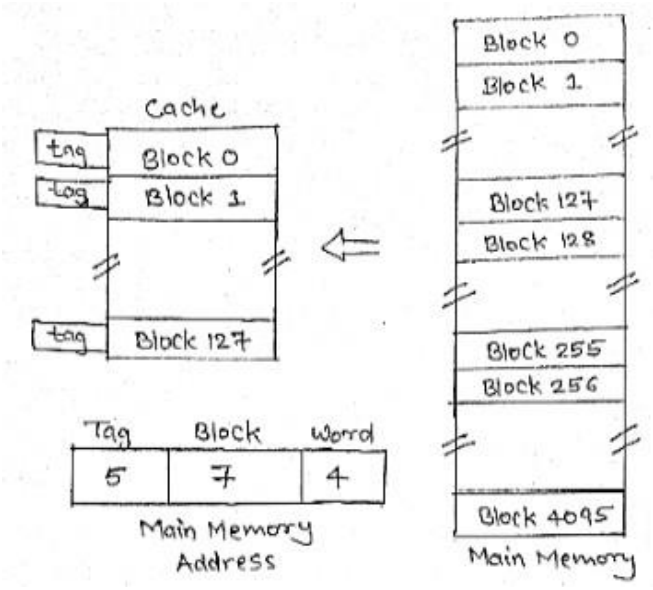
Cache Mapping Technique:-

The different Cache mapping techniques are as follows:-

1) Direct Mapping
2) Associative Mapping
3) Set Associative Mapping

Consider a cache consisting of 128 blocks of 16 words each, for total of 2048(2K) works and assume that the main memory is addressable by 16 bit address. Main memory is 64K which will be viewed as 4K blocks of 16 works each.

(1) Direct Mapping:-



1) The simplest way to determine cache locations in which store Memory blocks is direct Mapping technique.

2) In this block J of the main memory maps on to block J modulo 128 of the cache. Thus main memory blocks 0,128,256,….is loaded into cache is stored at block 0. Block 1,129,257,….are stored at block 1 and so on.

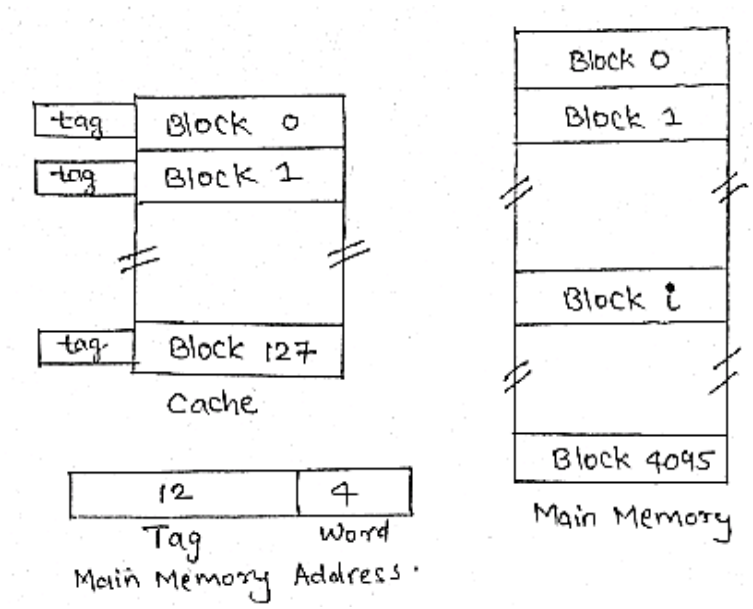
3) Placement of a block in the cache is determined from memory address. Memory address is divided into 3 fields, the lower 4-bits selects one of the 16 words in a block.

4) When new block enters the cache, the 7-bit cache block field determines the cache positions in which this block must be stored.

5) The higher order 5-bits of the memory address of the block are stored in 5 tag bits associated with its location in cache. They identify which of the 32 blocks that are mapped into this cache position are currently resident in the cache.

6) It is easy to implement, but not Flexible

(2) Associative Mapping:-



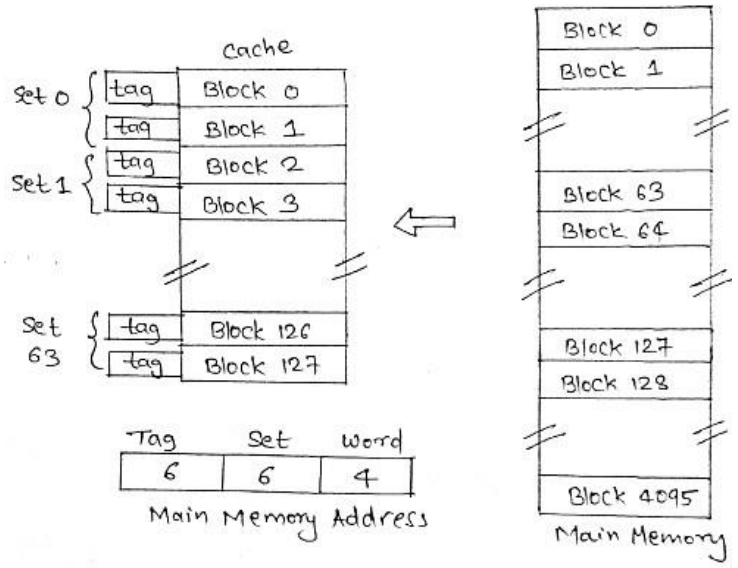
1) This is more flexible mapping method, in which main memory block can be placed into any cache block position.

2) In this, 12 tag bits are required to identify a memory block when it is resident in the cache.

3) The tag bits of an address recevied from the processor are compared to the tag bits of each block of the cache to see, if the desired block is present. This is known as Associative Mapping technique.

4) Cost of an associated mapped cache is higher than the cost of direct-mapped because of the need to search all 128 tag patterns to determine whether a block is in cache. This is known as associative search.

(3) Set-Associated Mapping:-



1) It is the combination of direct and associative mapping technique.

2) Cache blocks are grouped into sets and mapping allow block of main memory reside into any block of a specific set. Hence contention problem of direct mapping is eased , at the same time , hardware cost is reduced by decreasing the size of associative search.

3) For a cache with two blocks per set. In this case, memory block 0, 64, 128,…..,4032 map into cache set 0 and they can occupy any two block within this set.

4) Having 64 sets means that the 6 bit set field of the address determines which set of the cache might contain the desired block. The tag bits of address must be associatively compared to the tags of the two blocks of the set to check if desired block is present. This is two way associative search.

Virtual Memory

Virtual memory is a technique that allows the execution of processes which are not completely available in memory. The main visible advantage of this scheme is that programs can be larger than physical memory. Virtual memory is the separation of user logical memory from physical memory.

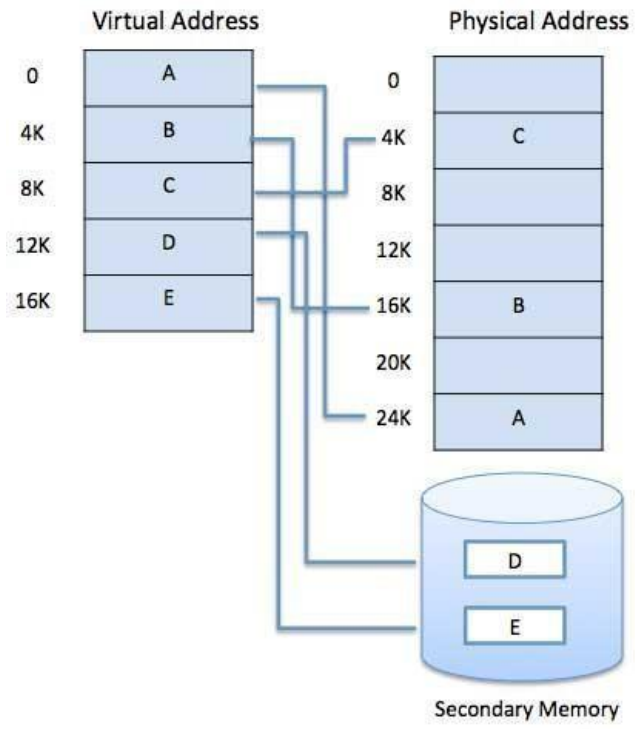
This separation allows an extremely large virtual memory to be provided for programmers when only a smaller physical memory is available. Following are the situations, when entire program is not required to be loaded fully in main memory.

- User written error handling routines are used only when an error occurred in the data or computation.
- Certain options and features of a program may be used rarely.
- Many tables are assigned a fixed amount of address space even though only a small amount of the table is actually used.
- The ability to execute a program that is only partially in memory would counter many benefits.
- Less number of I/O would be needed to load or swap each user program into memory.
- A program would no longer be constrained by the amount of physical memory that is available.
- Each user program could take less physical memory, more programs could be run the same time, with a corresponding increase in CPU utilization and throughput.

- A computer can address more memory than the amount physically installed on the system. This extra memory is actually called virtual memory and it is a section of a hard disk that's set up to emulate the computer's RAM.

Modern microprocessors intended for general-purpose use, a memory management unit, or MMU, is built into the hardware. The MMU's job is to translate virtual addresses into physical addresses.

A basic example is given below -



Auxiliary Memory

Auxiliary memory is much larger in size than main memory but is slower. It normally stores system programs, instruction and data files. It is also known as secondary memory.

It can also be used as an overflow/virtual memory in case the main memory capacity has been exceeded. Secondary memories cannot be accessed directly by a processor.

First the data/information of auxiliary memory is transferred to the main memory and then that information can be accessed by the CPU.

Characteristics of Auxiliary Memory are following −

- Non-volatile memory − Data is not lost when power is cut off.
- Reusable − The data stays in the secondary storage on permanent basis until it is not overwritten or deleted by the user.
- Reliable − Data in secondary storage is safe because of high physical stability of secondary storage device.
- Convenience − With the help of a computer software, authorised people can locate and access the data quickly.
- Capacity − Secondary storage can store large volumes of data in sets of multiple disks.
- Cost − It is much lesser expensive to store data on a tape or disk than primary memory.

Associative memory

Associative Memory comprises of a memory array and logic for m words with n bits per word.

Argument register A and key register K both have n bits, one for every bit of a word. Match register M has m bits, one for each memory word.

Every word in memory is compared in parallel with content of argument register then the words which match the bits of argument register set a corresponding bit in match register. After matching process those bits in match register which have been set denote the fact that their corresponding words have been matched.

Reading is achieved by a sequential access to memory for those words whose corresponding bits in match register have been set.

Key register offers a mask for choosing a specific key or field in argument word. The complete argument is compared with every memory word if key register contains all 1s. Or else only those bits in argument which have 1s in their corresponding positions of key register are compared. So the key offers a mask or identifying information that specifies how reference to memory is made.

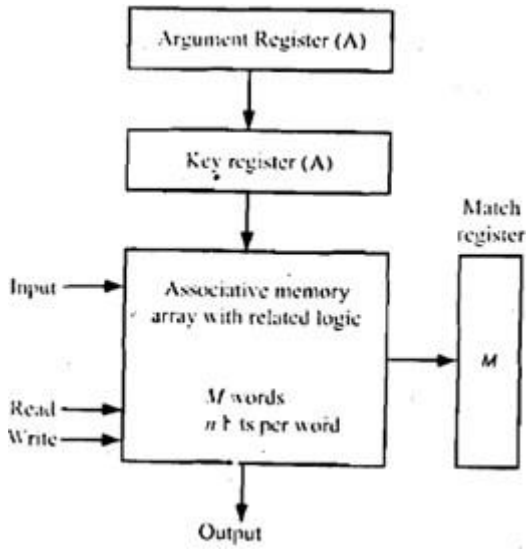


Fig: Associative Memory - Block Diagram

To explain with a numerical illustration assume that argument register A and key register K have the bit configuration displayed below. Only three leftmost bits of a compared with memory words since K has 1's on these positions.

A	101 111100	
K	111 000000	
Word 1	100 111100	no match
Word 2	101 000001	match

Word 2 matches unmasked argument field since the three leftmost bits of argument and word are equal.